CrystEngComm

Accepted Manuscript

View Article Online
DOI: 10.1039/C6CE01940D

CrystEngComm

ARTICLE

Received 00th January 20xx,
Accepted 00th January 20xx

DOI: 10.1039/x0xx00000x

www.rsc.org/

pH-controlled assembly of three-dimensional tungsten oxide hierarchical nanostructures for catalytic oxidation of cyclohexene to adipic acid

Lei Wang[a,b], Mengqiu Huang[a,b], Zhangxian Chen[a,b,*], Zeheng Yang[a,b,*], Maoqin Qiu[a,b], Kai Wang[a,b], Weixin Zhang[a,b,*]

Abstract: Tungsten oxide has wide applications in chromic device, catalysis, etc. An effective control over the morphology of tungsten oxide remains a major challenge. Herein, we present a facile pH-controlled strategy to synthesize three-dimensional tungsten oxide architectures with different morphologies without using any templates or surfactants. These hierarchical architectures with the rod-like, disk-like and sphere-like morphologies are assembled by the one-dimensional tungsten oxide nanowire/nanorods. The influence of single parameter of pH values on the morphologies and crystal structures of tungsten oxide is systematically studied. Results indicate that the different polytungstate anions obtained at various pH values give rise to different crystal nuclei by dehydration process. These crystal nuclei grow into the similar phases (crystal structure) but distinct morphologies via hydrothermal treatment. Here, the similarity of the crystal structures of h-WO_3 and o-WO_3·0.33H_2O play a key role in the structural transformation. Further experiments show that these three-dimensional tungsten oxide architectures display promising applications as the catalyst in green chemistry for the oxidation of cyclohexene to adipic acid by using the the environmentally friendly oxidant (hydrogen peroxide).

1. Introduction

The nanostructuring has emerged as an effective approach to unlock the potential of many functional materials and offer them special properties different from those of their bulk counterparts. As an important metal oxide, tungsten trioxide (WO_3), including its hydrates (WO_3·xH_2O, x=0–2), has attracted great research interest because of their unique properties and wide applications in chromic devices[1,2], photoluminescence[3], photocatalysis[4-6], gas sensing[7-9]. In addition, as a promising non-precious catalyst in the field of organic chemistry, tungsten oxide has been used in various catalytic systems, such as oxidation of cyclohexanol, cyclopentene, hydrogenation of thiophene[10-12]. As an effective support of precious metals, it also presents decent catalytic activities in oxidations of toluene and cyclohexene[13-15].

To meet the need for diverse applications, great effort has been devoted to synthesize tungsten oxide and its hydrates with controllable morphologies, including one-dimensional (1D) (e.g. nanorods[16], nanowires[17], and nanotubes[18,19]), two-dimensional (2D) (e.g. nanosheets[20,21], nanoribbons[22]) and three-dimensional (3D) nanostructures (e.g. urchin-like spheres[23], brush-like hierarchical structures[24], nanotrees[25], nanoflowers[26]). Among them, 3D hierarchical nanostructures have attracted more attention due to their porous nanostructures composed by the adjacent building blocks. These nanostructures with a large void space have high accessible surface area and a huge amount of available active sites, which can facilitate the diffusion and adsorption of the reactants during reactions. Thus, these hierarchical nanostructures are considered as feasible candidates for gas sensors, photo catalysts and organic catalysts.

Some advancement has been made in fabricating 3D tungsten oxide nanostructures with enhanced performances. For example, Xu et al.[27] prepared the flower-like and wheel-like tungsten oxide architectures consisting of rod-like building blocks, by adjusting the pH values of the precursor solutions using NaCl as the structure-construction agent. As the pH value was 2.0, the wheel-like tungsten oxide architectures were synthesized, whereas the flower-like morphology can be obtained at the pH value of 1.0. These WO_3 architectures showed high photocatalytic activity towards the degradation of rhodamine B under ultraviolet light irradiation. Li et al.[28] proposed an aqueous approach to synthesize metastable hexagonal tungsten oxide (h-WO_3) hierarchical nanostructures using polyvinyl pyrrolidone (PVP) as a soft template. They found that the formation of WO_3 crystallites was dramatically influenced by the pH values. The lower pH values led to a faster reaction, where the nucleation and growth of the

[a] School of Chemistry and Chemical Engineering, Hefei University of Technology, Hefei, China 230009. E-mail: wxzhang@hfut.edu.cn (W.X. Zhang).
[b] Anhui Key Laboratory of Controllable Chemical Reaction & Material Chemical Engineering, Hefei, China 230009.

Electronic Supplementary Information (ESI) available: FTIR spectra, FESEM images, TEM image and XRD pattern of the as-prepared tungsten oxide and commercial WO_3. Reaction scheme for the condensation of tungstate ions. Plausible pathway for the catalytic oxidation of cyclohexene to adipic acid. See DOI: 10.1039/x0xx00000x

View Article Online
DOI: 10.1039/C6CE01940D

metastable phase were accelerated. The hexagonal tungsten oxide showed excellent recycled visible-light photochromic performance. Li et al.[29] synthesized hierarchical tungsten oxide nanostructures assembled by one-dimensional nanosized building blocks. The tungsten oxide nanostructures with controllable phases and exposed facets can be obtained at different pH values (0.74, 1.08, 6.80) adjusted using urea. The o-$WO_3 \cdot 0.33H_2O$ exposing the largest amount of (020) facets showed the highest photocatalytic activity towards the degradation of rhodamine B under visible light illumination. Salmaoui et al.[30] reported the synthesis of o-$WO_3 \cdot 0.33H_2O$ with chrysanthemum-like and disk-like architectures by surfactant-assisted hydrothermal reaction in acidic medium (pH=1). N-alkyl chain length of the surfactant [$C_nH_{(2n+1)}SO_4Na$ (with n=10, 12 and 14)] was found to play a key role in the crystallite sizes and the morphology. The as-prepared samples displayed reversible redox behavior in $LiClO_4$/propylene carbonate accompanied by reversible intercalation/ deintercalation of Li^+ cations into their crystal matrices. Chen et al.[6] reported the synthesis of hierarchical WO_3 hollow shells with dendritic, spherical and dumbbell-like shapes by calcining the acid-treated $PbWO_4$ and $SrWO_4$ templates. The hierarchical WO_3 nanostructures displayed enhanced photocatalytic activities for the degradation of rhodamine B under simulated visible light. Gu et al.[23] synthesized WO_3 with the urchin-like and ribbon-like nanostructures, via a hydrothermal approach, which showed good electrochromic properties. It was found that the morphologies of the final products can be controlled using different sulfates as the additives. The urchin-like WO_3 can be produced by using Rb_2SO_4 as the additive, whereas the ribbon-like morphology was obtained using K_2SO_4. They proposed that the ribbon-like WO_3 nanostructures were assembled by the wire-shape building blocks via a sulfate-induced oriented attachment growth mechanism.

Although varieties of methods have been proposed to synthesize the hierarchical 3D WO_3 nanostructures and its hydrates, the reported approaches generally require sacrificial templates (e.g. PAL2-16[5], PEO-b-PS[9], $PbWO_4$ and $SrWO_4$[6]), surfactants (e.g. PVP[28], SDS[30]) and organic or inorganic additives (e.g. $(NH_4)_2Fe(SO_4)_2 \cdot 6H_2O$[31], $C_2H_2O_4 \cdot 2H_2O$ and Rb_2SO_4[23]). More importantly, an effective control over the morphology of the final products is still a major challenge. Therefore, it is highly desirable to develop a facile strategy for the fabrication of 3D WO_3 and its hydrate nanostructures with well-defined morphologies.

Adipic acid is an important industrial chemical intermediate mainly used in the manufacture of Nylon-6,6, for synthetic fibers, polyurethanes, plasticizers and adiponitriles. The current industrial production of adipic acid mainly employs the oxidation method in which nitric acid oxidizes the cyclohexanone/cyclohexanol mixture, so-called KA oil. However, this method is not preferred due to the emission of environmentally hazardous gas N_2O, the corrosion of reaction vessels and the energy-intensive process. In view of this, catalytic oxidation of cyclohexene exists as a better alternative to produce adipic acid. To further improve the catalytic reaction, various reaction systems or media have been developed, including H_2O_2 and Na_2WO_4 catalyst[32,33], silica supported 1-Butyl-3-methylimidazolium tungstate ($[BMIm]_2WO_4$) ionic liquid[34], dual task-specific ionic liquid[35], $H_2WO_4/H_2SO_4/H_3PO_4$ mixture[36], silica-functionalized ammonium tungstate catalyst[37], silver tungstate nano-rods catalyst in a Brønsted acidic ionic liquid[38]. Therefore, it is highly desirable to develop a green synthesis of adipic acid using environmentally benign catalytic oxidation method.

Herein, we present a facile pH-controlled strategy to synthesize WO_3 and its hydrates with different morphologies by simply adjusting the pH of the precursor solutions. The synthetic recipes only require the two reactants without using any templates or surfactants. The tungsten trioxide nanostructures with rod-like, disk-like and sphere-like architectures, based on the self-assembly of one-dimensional nanowires/nanorods, can be obtained by simply adjusting the pH of the precursor solutions. The variations of the morphologies and crystal structures with the pH values are also discussed. Results indicate that the different polytungstate anions obtained at various pH values give rise to different crystal nuclei by dehydration process. These crystal nuclei grow into the similar phases (crystal structure) but distinct morphologies via hydrothermal treatment. Here, the similarity of the crystal structures of h-WO_3 and o-$WO_3 \cdot 0.33H_2O$ play a key role in the structural transformation. Furthermore, these tungsten oxide nanostructures were used as the catalyst for the oxidation of cyclohexene by green oxidant (hydrogen peroxide) to produce adipic acid. These catalysts display high yields of adipic acid up to 74.4%. This work may pave the way for the synthesis of tungstic oxide with different nanostructures using a facile and controllable approach.

2. Experimental

2.1 Synthetic procedures

All the reagents (Analytical grade) were used without any further purification. Tungsten oxide nanostructures with different morphologies were synthesized via a hydrothermal process. In a typical procedure, 30 mL of acetic acid solution with pH=2 was prepared. Then, 0.3 g of $Na_2WO_4 \cdot 2H_2O$ was added and stirred for 30 min. The final pH of the mixed solution was adjusted to different values using HCl solution (3M). The solution was transferred into a 50 mL Teflon-lined stainless-steel autoclave and maintained at 180 ^oC for 24 h. The precipitate was centrifuged, washed with ethanol and deionized water, and finally dried at 60 ^oC for 10 h. It was found that no product can be obtained when the final pH is larger than 4.0. We prepared tungsten oxide nanostructures at different final pH values of 0.5, 1, 1.5, 2.0, 2.5, and 3.0, denoted as WO_3-0.5, WO_3-1.0, WO_3-1.5, WO_3-2.0, WO_3-2.5 and WO_3-3.0, respectively.

2.2 Characterization

The crystal structure was characterized by X-ray diffraction (XRD, Rigaku D/max-γB) with a Cu K_α radiation source

View Article Online
DOI: 10.1039/C6CE01940D

(λ=0.15418 nm) operated at 40 kV and 80 mA. The morphologies and structures were characterized using field-emission scanning electron microscope (FESEM, Hitachi-SU8020) and high-resolution transmission electron microscopic (HRTEM, JEM-2100F) at an accelerating voltage of 5 kV and 200 kV, respectively. N_2 adsorption-desorption isotherms were recorded on a Quantachrome NOVA 2200e surface area and pore size analyzer at liquid nitrogen temperature. The specific surface area was calculated by the multipoint Brunauer–Emmett–Teller (BET) procedure. The pore diameter and pore volume were calculated based on the Barrett–Joyner–Halenda (BJH) method. ^{1}H and ^{13}C NMR nuclear magnetic resonance (NMR) spectra were recorded on a VNMRS600 NMR spectrometer with deuterated DMSO as the solvent. Fourier transform infrared (FTIR) spectra were acquired on a Shimadzu JP FTIR spectrometer, using KBr as the reference.

2.3 Synthesis of adipic acid

The as-prepared tungsten oxide nanostructures were used as the catalyst for the synthesis of adipic acid. A typical reaction was carried out in a 25 mL three-necked round bottom flask equipped with a reflux condenser and a magnetic stirrer. As-prepared WO_3 sample (0.289 g, 1.25 mmol), oxalic acid (0.113 g, 1.25 mmol) and aqueous hydrogen peroxide (30 wt%, 22.5 mL) were first added into the flask with vigorous agitation. After 15 minutes, cyclohexene (5.25 mL, 50 mmol) was added into the above solution. The mixture was heated at 73 ^{o}C for 2 h and then at 90 ^{o}C for 18 h. The resultant aqueous mixture was cooled at 5 ^{o}C for 12 h. A white crystalline solid (adipic acid) was filtered off. For comparison, the catalytic properties of manganese oxide nanorods (synthesized according to our work[39,40]) were also tested.

3. Results and discussion

3.1 Structure and morphology

Fig. 1 shows the X-ray diffraction patterns for WO_3 and its hydrates synthesized at different pH values. All the diffraction peaks of WO_3-2.5 and WO_3-3.0 can be indexed to hexagonal WO_3 (h-WO_3, JCPDS card no.: 75-2187). No obvious impurities can be detected. When the pH value decreases to 2.0, the sample is characterized as orthorhombic $WO_3{\cdot}0.33H_2O$ (o-$WO_3{\cdot}0.33H_2O$, JCPDS card no.: 72-0199), which is confirmed by the FTIR data of the $WO_3{\cdot}0.33H_2O$ and WO_3 (**Fig. S1**). The broad absorption peaks between 1000 cm^{-1} and 600 cm^{-1} are characteristic of the stretching vibrations of W=O bonds and the O−W−O stretching vibrations. The peaks at 1620 cm^{-1} and 3455 cm^{-1} for $WO_3{\cdot}0.33H_2O$ are attributed to the scissoring and stretching mode of O−H, respectively[30].

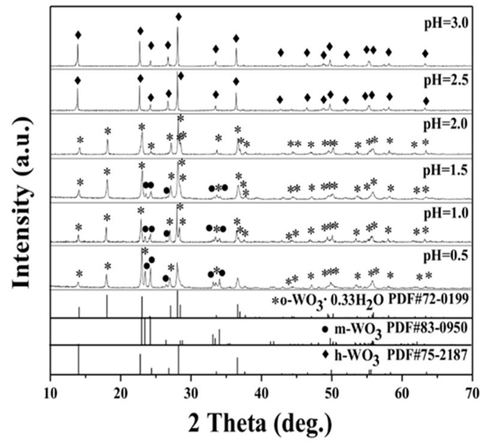

Fig. 1 XRD patterns of WO_3 and its hydrates synthesized at different final pH values: 0.5, 1.0, 1.5, 2.0, 2.5 and 3.0, respectively.

A further decrease of pH to 1.5, 1.0 and 0.5 results in the mixture of o-$WO_3{\cdot}0.33H_2O$ as the major component and monoclinic WO_3 (m-WO_3, JCPDS card no.: 83-0950) as the minor one. A typical experiment with final pH=1.0 was conducted to investigate the crystal evolution as a function of time. As shown in **Fig. 2**, the sample can be identified as o-$WO_3{\cdot}0.33H_2O$ at a reaction time of 1 h. Prolonging the reaction time to 12 h leads to the mixture of m-WO_3 and o-$WO_3{\cdot}0.33H_2O$, while the crystallinity slightly increases as the reaction time is extended.

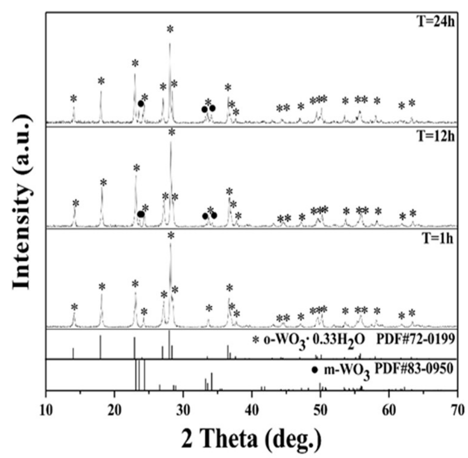

Fig. 2 XRD patterns of WO_3 microspheres at pH=1 with different hydrothermal times: 1 h, 12 h and 24 h.

Our experiments indicate that the crystal structures of the tungsten oxide are highly related to the pH values of the precursor solution. It was reported that h-WO_3 can be obtained by the complete dehydration of polytungstate[27]. Therefore, it is understandable that a slow reaction system (high pH value) leads to the complete removal of water molecules to form h-WO_3 under hydrothermal conditions. With the decrease of pH, the more amount of H^+ lead to a faster reaction, resulting in the incomplete dehydration of

View Article Online
DOI: 10.1039/C6CE01940D

polytungstate. Thus, o-$WO_3 \cdot 0.33H_2O$ crystal was finally obtained.

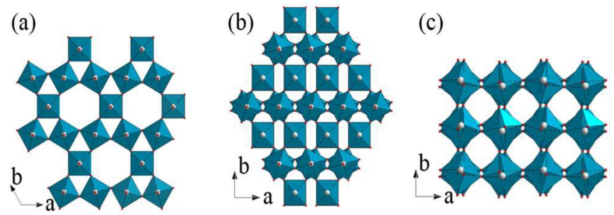

Fig. 3 Simulated crystal structures of (a) h-WO_3, (b) o-$WO_3 \cdot 0.33H_2O$ and (c) m-WO_3 crystals. Color codes: white (W atom), red (O atom).

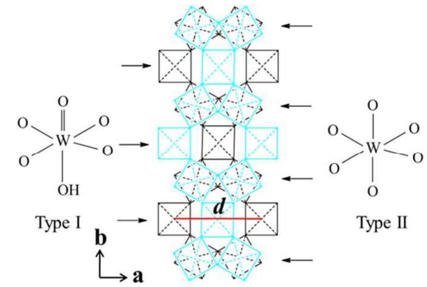

Fig. 4 Schematic illustration of adjacent layers of the $WO_3 \cdot 0.33H_2O$ structure along the [001] axis.

To understand the structural transformation, the simulated structures of o-$WO_3 \cdot 0.33H_2O$ and h-WO_3 are presented. As shown in **Fig. 3a**, the h-WO_3 consists of WO_6 octahedral structures sharing the equatorial oxygen in the a, b plane of WO_6. The complete overlapping of WO_6 along c axis leads to the formation of trigonal and hexagonal tunnels. The crystal structure of o-$WO_3 \cdot 0.33H_2O$ is shown in **Fig. 3b** and also in **Fig. 4**. The structure consists of two different octahedra. For type I octahedra [$WO_5(H_2O)$], the two oxygen atoms form the W=O bond and the longer W-OH_2 bond, respectively. For type II octahedral [WO_6], all W-O bonds share the same bond length. Two different octahedra form the first layer by sharing the equatorial oxygen in a, b planes (black layer in **Fig. 4**). Different from h-WO_3, the second layer (blue layer in **Fig. 4**) in o-$WO_3 \cdot 0.33H_2O$ is shifted by $d/2$ perpendicular to the c axis. The similarity of the crystal structures facilitates the transformation between h-WO_3 and o-$WO_3 \cdot 0.33H_2O$. **Fig. 3c** displays the structure of m-WO_3, which similarly consists of WO_6 octahedra connected by sharing corner oxygen atoms. The difference is that the WO_6 octahedra are slightly distorted and tilted.

Both the structures of h-WO_3 and m-WO_3 consist of WO_6 octahedra but with different arrangements. As reported in the literature[41], o-$WO_3 \cdot 0.33H_2O$ has a free energy higher than that of m-WO_3. It is thermodynamically favorable that the initially formed o-$WO_3 \cdot 0.33H_2O$ transforms into m-WO_3. This could explain the co-existence of the two phases after longer reaction durations (**Fig. 2**).

Fig. 5 presents the FESEM images of the tungsten oxide synthesized at the corresponding pH values. It is clear that the pH value is crucial in controlling the morphology of tungsten oxide. As displayed in **Fig. 5a** and **b**, the porous microspheres (WO_3-1.0) containing a large number of nanorods packing together can be obtained when pH value is 1.0. The sample of WO_3-2.0 is formed by the microdisks with diameter around 3-5 μm and thickness around 1 μm (**Fig. 5c** and **d**). The microdisks are essentially assembled by the closely packed tungsten oxide nanorods (**Fig. S2**). When the pH value further increases to 2.5, the nanorod bundles of 8–12 μm in length can be obtained (**Fig. 5e** and **f**). Further analysis of these bundles suggests their highly oriented hierarchical nanostructures (**Fig. S3**). A further increase of pH value to 3.0 leads to the monodispersed rod-like structures of h-WO_3 with diameters of 200–300 nm and length of 8–15 μm are obtained (**Fig. 5g** and **h**).

Here, WO_3-1.0 was chosen to study the formation process of tungsten oxide microspheres as a function of reaction time. As shown in **Fig. 6a** and **b**, the urchin-like microspheres with the size around 2 μm covered by small nanorods can be obtained after a hydrothermal reaction for 2 h. Increasing the reaction time to 8 h gives rise to larger particle size and thicker nanorods packing on the surface (**Fig. 6c** and **d**). Further increase of the reaction time to 24 h results in larger microspheres with diameters of several microns (**Fig. 6e** and **f**) and denser nanorods packing on the surface (**Fig. S4**).

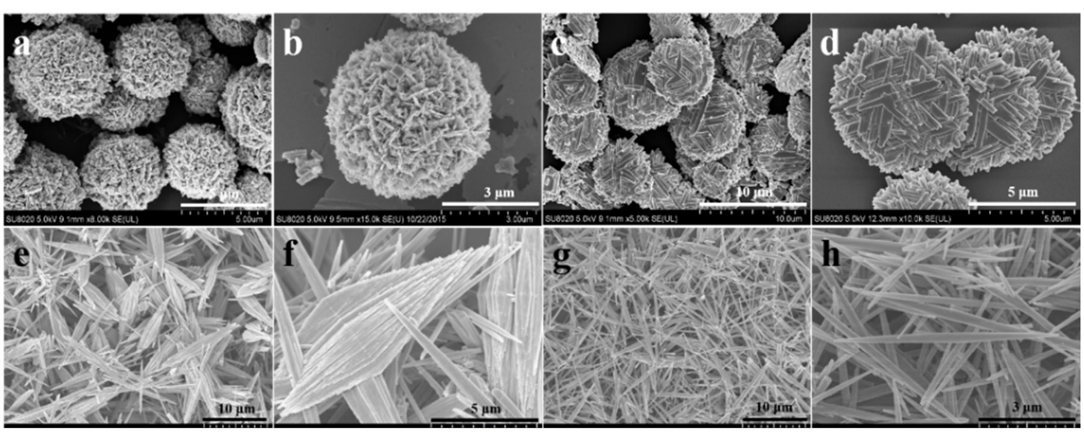

Fig. 5 FESEM images of the different samples: (a, b) WO_3-1.0, (c, d) WO_3-2.0, (e, f) WO_3-2.5 and (g, h) WO_3-3.0.

View Article Online
DOI: 10.1039/C6CE01940D

CrystEngComm

ARTICLE

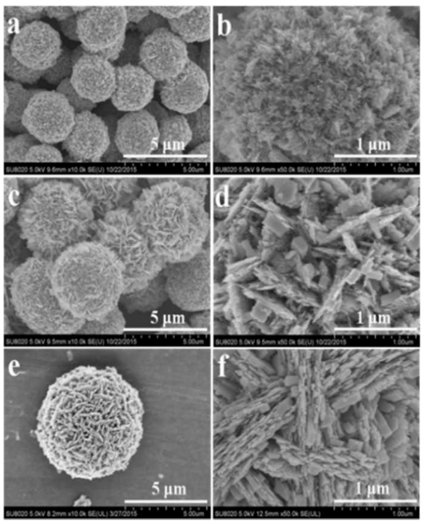

Fig. 6 Morphological evolution of WO_3-1.0 synthesized at different hydrothermal times: (a, b) 2 h, (c, d) 8 h, and (e, f) 24 h.

The TEM image of tungsten oxide microsphere (WO_3-1.0) is shown in **Fig. 7a**. The HRTEM image (**Fig. 7b**) shows the lattice spacings of 0.260 nm and 0.321 nm, corresponding to the (202) plane of m-WO_3 and (220) plane of WO_3•$0.33H_2O$, respectively. The result implies that the microsphere is the mixture of m-WO_3 and o-WO_3•$0.33H_2O$, in good consistence with the XRD result. **Fig. 7c** show the TEM image of $WO_3 \cdot 0.33H_2O$ microdisks (WO_3-2.0) that is composed of nanorods. The spacings of the lattice fringes are found to be about 0.321 nm and 0.331 nm, respectively, corresponding to the (220) and (131) planes of orthorhombic $WO_3 \cdot 0.33H_2O$ (**Fig. 7d**). **Fig. 7e** presents the TEM image of a typical h-WO_3 nanorod (WO_3-3.0), which confirms that each rod is assembled by a bundle of closely packed and highly aligned h-WO_3 nanowires with the diameters of about 10 nm. The corresponding HRTEM image (**Fig. 7f**) clearly displays the ordered atomic arrangement in h-WO_3 nanorods. The lattice spacings of 0.390 nm and 0.370 nm can be identified to the (001) and (110) planes of hexagonal WO_3, respectively, with the nanorod growing parallel to the [001] direction.

The results indicate that the tungsten oxide samples have significantly different morphologies though they were prepared under similar conditions except the different pH values. According to the references[42,43], polytungstate anions are sensitive to the pH of the precursor solution (**Fig. S5**). The monotungstate ion (WO_4^{2-}) only exists at pH larger than 9. With the pH ranging from 5 to 9, paratungstate A $[W_7O_{24}]^{6-}$ and paratungstate B $[H_2W_{12}O_{42}]^{10-}$ ions coexist in the solution.

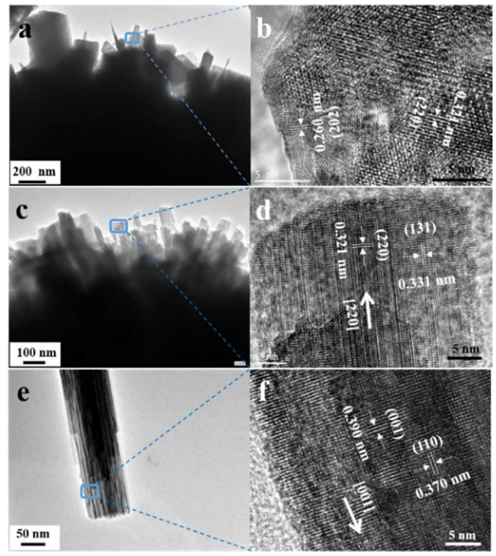

Fig. 7 (a, c, e) TEM and (b, d, f) HRTEM images of different samples: (a, b) WO_3-1.0 ($WO_3 \cdot 0.33H_2O$ microspheres), (c, d) WO_3-2.0 ($WO_3 \cdot 0.33H_2O$ microdisks) and (e, f) WO_3-3.0 (h-WO_3 nanorods).

When the pH value is between 2 and 4, paratungstate A and paratungstate B ions transform into the metatungstate $[H_2W_{12}O_{40}]^{6-}$, also called Keggin structure. At pH range of 1–2, tungstate Y $[W_{10}O_{32}]^{4-}$ anions form in the solution.

Fig. 8 illustrates the growth mechanism of tungsten oxide with different morphologies. For WO_3-3.0, the $[H_2W_{12}O_{40}]^{6-}$ anions formed in the precursor solution decompose into WO_3 nuclei during the following hydrothermal treatment. Acetic acid (CH_3COOH) can form chelates with tungsten species by -O-W-O- linkage in aqueous medium, thus to prevent the aggregation of the nuclei and kinetically control the growth rate of different crystal faces through selective adsorption and desorption.[44] These WO_3 nuclei, serving as the seeds, grow along [001] direction to form small wires. Continuous growth consumes the small wires to form long and uniform nanowires, due to the well-known Ostwald ripening process. Simultaneously, the self-assembly of these nanowires occurs to minimize the surface energies via formation of the h-WO_3 nanorods. For the sample prepared at pH of 2.5, the more acetic acid molecules form more chelates with tungsten species, preventing further aggregation of the nanowire into nanorod. Therefore, the nanorods with small radius can be obtained, which subsequently form nanorod bundles to minimize their surface energy. Different from the Keggin structure, the $[W_{10}O_{32}]^{4-}$ anions are formed at the pH of 2.0, which decompose into $WO_3(H_2O)_{0.33}$ nuclei via incomplete

View Article Online
DOI: 10.1039/C6CE01940D

dehydration upon hydrothermal treatment. Similar to the growth process of WO_3-3.0, $WO_3(H_2O)_{0.33}$ nanorods are formed via the oriented growth from the nuclei along with [220] direction. The final disk shape can be obtained from the self-assembly of these nanorods. This result is consistent with the reported work.[45] For the growth process of WO_3-1.0, the $[W_{10}O_{32}]^{4-}$ anions decompose into $WO_3(H_2O)_{0.33}$ nuclei. In higher concentrations of H^+ than other precursor solution, the crystallites produced at fast nucleation process spontaneously aggregate into sphere-like architectures to reduce surface energy via hydrothermal reaction (**Fig. S6**). Then, a secondary nucleation occurs at the surface defect sites of microsphere, which could provide high energy for the growth of nanocrystals.[46] Meanwhile, CH_3COO^- anions, adsorbed selectively onto specific faces of the freshly formed secondary crystals, reduce the energy of the system. Thus, the newly formed particles grow into nanorods around the microsphere due to the anisotropic growth. Finally，these microspheres are consisted of smooth nanorods, which is similar to the disc-like nanostructures of WO_3-2.0.

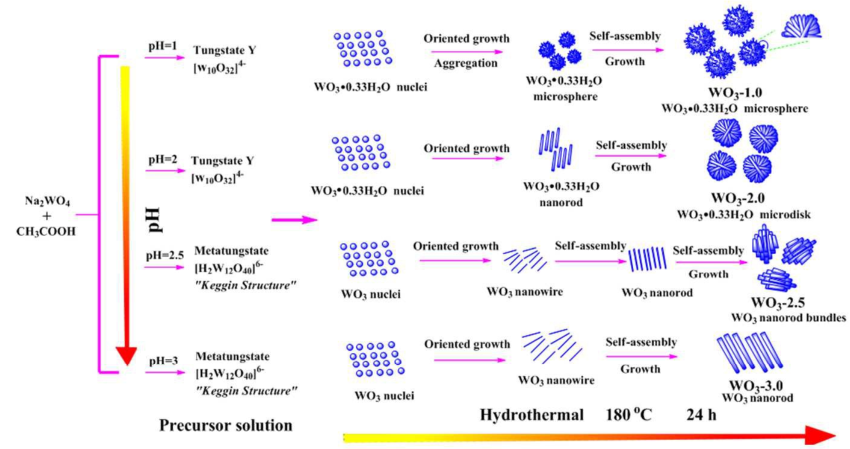

Fig. 8 Schematic illustration of the growth process of tungsten oxide with different morphologies.

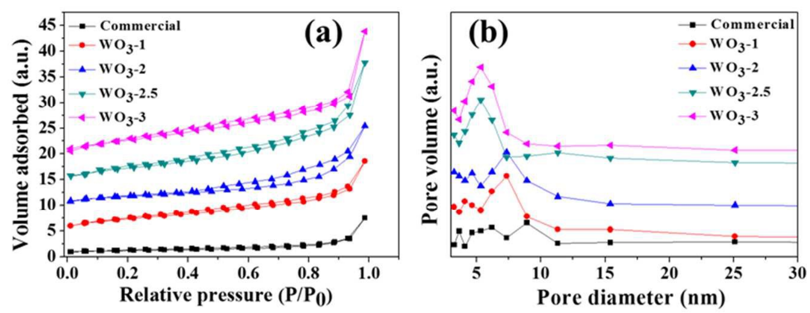

Fig. 9 (a) The nitrogen adsorption-desorption isotherms, and (b) the BJH pore size distributions of WO_3-1.0, WO_3-2.0, WO_3-2.5, WO_3-3.0 and commercial WO_3.

Table 1 The porous structure parameters of different samples

Sample	Specific surface area (m^2/g)	Average pore size (nm)	Pore volume (cm^3/g)
Commercial WO_3	1	8.9	0.010
WO_3-1.0	13	7.3	0.058
WO_3-2.0	10	7.2	0.045
WO_3-2.5	14	5.0	0.057
WO_3-3.0	17	5.1	0.065

It is noted that WO_3-1.0, WO_3-2.0, WO_3-2.5 contain many microscopic pores and WO_3 nanorods stack together while leaving porous microstructures among the rods. The nitrogen adsorption-desorption isotherms are used to characterize these tungsten oxide nanostructures. The adsorption and desorption behaviors of these tungsten oxide, displayed in **Fig. 9a**, show typical type-III isotherms with hysteresis loop which is typical for mesoporous materials. Owing to the weak adsorbent-adsorbate interaction, adsorbate show a low uptake at the low pressure and a rapid increase of uptake at elevated pressures. The commercial WO_3 exhibits similar adsorption-desorption isotherm but with a relatively lower uptake of the adsorbate. **Table 1** lists the BET surface area, average pore size and volume of WO_3-1.0, WO_3-2.0, WO_3-2.5, WO_3-3.0 and commercial WO_3. Apparently, all these tungsten oxide nanostructures have a much higher surface area, larger pore volume and narrower pore size distribution than the commercial one. As revealed by **Fig. S7**, the commercial WO_3 shows large particle sizes up to hundreds microns, leading to a much lower surface area. With large surface area and porous

View Article Online
DOI: 10.1039/C6CE01940D

structure, these tungsten oxide catalysts can effectively diffuse and accessible to reactants.

3.2 Catalytic performance

The as-prepared tungsten oxide nanostructures were used as the catalyst in the selective oxidation of cyclohexene to adipic acid at 90 ^oC under ambient pressure using environmentally friendly oxidant of aqueous hydrogen peroxide. The melting point range was measured to be 151 ^oC–152 °C for the synthesized adipic acid. The FTIR, ^1H and ^{13}C NMR spectra shown in **Fig. S8** clearly confirm the obtained adpic acid. As tabulated in **Table 2**, hydrogen peroxide does not show any oxidative activity towards cyclohexene (Entry 1 in **Table 2**). Here, the tungsten oxide nanostructures with different morphologies can effectively catalyze the oxidation of cyclohexene by H_2O_2, resulting in high yields of 74.4 %, 61.0 %, 60.3 % and 65.8 %, respectively (Entry 6-9 in **Table 2**), significantly higher than that obtained by commercial WO_3 (Entry 5 in **Table 2**). Although these catalysts can also provide a large surface, it may not be completely accessible to the reactants, implying that the effective surface is much smaller than the tested value.[47] WO_3-1.0 may have the largest effective surface, thus displaying the highest yield of adipic acid. The mechanistic understanding on the different catalytic activities requires further investigations.

It has also been reported that manganese oxide nanostructures (e.g. MnO_2 and Mn_3O_4 nanorods) can greatly promote the decomposition of hydrogen peroxide by the catalytic effect to produce various active species, including atomic oxygen and radicals of ▪O, ▪OOH, which show strong oxidative activity towards some organics[48]. However, the addition of these nanorods does not show any catalytic activity towards cyclohexene (Entry 2-4 in **Table 2**). This implies that the catalytic oxidation of cyclohexene by tungsten oxide and H_2O_2 might be based on a different catalytic mechanism, which will be discussed later.

Table 2 The performance of different catalysts for synthesis of adipic acid

Entry	Catalyst	Yield of adipic acid (wt%)
1	None	-
2	α-MnO_2	-
3	β-MnO_2	-
4	Mn_3O_4	-
5	Commercial WO_3	31.9
6	WO_3-1.0	74.4
7	WO_3-2.0	61.0
8	WO_3-2.5	60.3
9	WO_3-3.0	65.8
10	Na_2WO_4	90[a]
11	$H_2WO_4/H_2SO_4/H_3PO_4$	94.1[b]
12	ammonium tungstate	84[c]
13	Ag_2WO_4	85[d]

Reaction conditions: cyclohexene (5.25 mL); H_2O_2 (22.5 mL, 30 wt%); oxalic acid (1.25 mmol); catalyst (1.25 mmol). No adipic acid products can be obtained for Entry 1-4.[a]: Ref. [33]. [b]: Ref. [36]. [c,d]: Ref. [37,38].

3.3 Catalytic mechanism

It was reported that the catalytic decomposition of hydrogen peroxide by manganese oxide nanostructures resulted in the formation of oxygen bubbles.[47,48] However, it was noted that no obvious bubbles can be observed when mixing hydrogen peroxide with these tungsten oxide catalysts. In addition, manganese oxides here did not show any catalytic activity towards the oxidation of cyclohexene (Entry 2-4 in **Table 2**). We speculate that hydrogen peroxide might react with the surface of tungsten oxide nanostructures to form an active complex of $H_2[WO(O_2)_2(OH)_2]$ which oxidizes cyclohexene to form adipic acid. The formation of the active complex may occur as following:[49, 50]

$$WO_3 + H_2O_2 \xrightarrow{H_2O} HOW(O)_2\text{-}OOH \quad (1)$$

$$HOW(O)_2\text{-}OOH + H_2O_2 \xrightarrow{H_2O} H_2\,[WO(O_2)_2(OH)_2] \quad (2)$$

According to the references[49,50], the catalytic oxidation of cyclohexene with aqueous hydrogen peroxide (30 wt%) as the oxidant might be based on multi-step reactions, including the oxidation, hydrolysis and condensation reactions (**Fig. S9**). Several intermediates exist during the whole process of the catalytic reaction, such as 1,2-epoxycyclohexane, 1,2-cyclohexanediol and adipic anhydride.

4 Conclusions

Tungstic oxide nanostructures with different morphologies have been successfully synthesized via a facile pH-controlled approach without using any templates or surfactants. The rod-like, disk-like and sphere-like 3D hierarchical architectures of tungsten oxide can be obtained by simply controlling the pH of the precursor solution. These 3D architectures are formed by the self-assembly of one-dimensional nanowire/nanorods of tungsten oxide and its hydrates. Experimental investigations indicate that the different polytungstate anions obtained at various pH values give rise to different crystal nuclei by dehydration process. These crystal nuclei grow into the similar phases (crystal structure) but distinct morphologies via hydrothermal treatment. Further experiments show that tungsten oxide has excellent potential applications in green chemistry for the synthesis of adipic acid by using the environmentally friendly oxidant of hydrogen peroxide. We believe that these rod-like, disk-like and sphere-like 3D hierarchical architectures of tungsten oxide can be applied in many fields, such as photocatalysis, gas sensor, owing to the large void and plenty of active surface sites.

Acknowledgements

The authors gratefully acknowledge the financial support from the National Natural Science Foundation of China (NSFC grants: 91534102, 21176054 and 21271058), Science and Technology Project of Anhui Province (1501021013) and Intelligent Manufacturing Institute of Hefei University of Technology (IMICZ2015104).

View Article Online
DOI: 10.1039/C6CE01940D

Notes and references

1 S. T. Wang, X. J. Feng, J. N. Yao and L. Jiang, *Angew. Chem., Int. Ed.*, 2006, **45**, 1264-1267.
2 G. A. Niklasson and C. G. Granqvist, *J. Mater. Chem. A*, 2007, **17**, 127-156.
3 M. Feng, A. L. Pan, H. R. Zhang, Z. A. Li, F. Liu, H. W. Liu, D. X. Shi, B. S. Zou and H. J. Gao, *Appl. Phys. Lett.*, 2005, **86**. 141901.
4 J. A. Seabold and K. S. Choi, *Chem. Mater.*, 2011, **23**, 1105-1112.
5 D. Chandra, K. Saito, T. Yui and M. Yagi, *Angew. Chem., Int. Ed.*, 2013, **52**, 12606-12609.
6 D. Chen and J. H. Ye, *Adv. Funct. Mater.*, 2008, **18**, 1922-1928.
7 X. Q. An, J. C. Yu, Y. Wang, Y.M. Hu, X. L. Yu and G. J. Zhang, *J. Mater. Chem.*, 2012, **22**, 8525-8531.
8 H. G. Moon, S. D. Han, M. G. Kang, W. S. Jung, B. Kwon, C. Kim, T. Lee, S. Lee, S. H. Baek, J. S. Kim, H. H. Park and C. Y. Kang, *Sensor Actuat. B-Chem.*, 2016, **229**, 92-99.
9 Y. H. Li, W. Luo, N. Qin, J. P. Dong, J. Wei, W. Li, S. S. Feng, J. C. Chen, J. Q. Xu, A. A. Elzatahry, M. H. Es-Saheb, Y. H. Deng and D. Y. Zhao, *Angew. Chem., Int. Ed.*, 2014, **53**, 9035-9040.
10 L. Wang, Z. X. Chen, M. Q. Huang, Z. H. Yang, P. Sun, K. Wang and W. X. Zhang, *Catal. Lett.*, 2016, **146**, 1283-1290.
11 Y. Xi, Z. Chen, V. Gan Wei Kiat, L. Huang and H. Cheng, *Phys. Chem. Chem. Phys.*, 2015, **17**, 9698-9705.
12 R. Jin, H. Li and J.F. Deng, *J. Catal.*, 2001, **203**, 75-81.
13 D. N. Durgasri, T. Vinodkumar, P. Sudarsanam and B. M. Reddy, *Catal. Lett.*, 2014, **144**, 971-979.
14 S. Ghosh, S. S. Acharyya, D. Tripathi and R. Bal, *J. Mater. Chem. A*, 2014, **2**, 15726-15733.
15 S. Ghosh, S. S. Acharyya, S. Adak, L. N. S. Konathala, T. Sasaki and R. Bal, *Green Chem.*, 2014, **16**, 2826-2834.
16 Z. J. Gu, Y. Ma, W. S. Yang, G. J. Zhang and J. N. Yao, *Chem. Commun.*, 2005, **28**, 3597-3599.
17 A. Phuruangrat, D. J. Ham, S. J. Hong, S. Thongtem and J. S. Lee, *J. Mater. Chem.*, 2010, **20**, 1683-1690.
18 Z. G. Zhao and M. Miyauchi, *Angew. Chem., Int. Ed.*, 2008, **47**, 7051-7055.
19 J. Li, X. H. Liu, Q. F. Han, X. X. Yao and X. Wang, *J. Mater. Chem. A*, 2013, **1**, 1246-1253.
20 Y. Hou, F. Zuo, A. P. Dagg, J. K. Liu and P. Y. Feng, *Adv. Mater.*, 2014, **26**, 5043-5049.
21 W. Xiao, W. T. Liu, X. H. Mao, H. Zhu and D. H. Wang, *J. Mater. Chem. A*, 2013, **1**, 1261-1269.
22 C. Lian, X. L. Xiao, Z. Chen, Y. X. Liu, E. Y. Zhao, D. S. Wang, C. Chen and Y. D. Li, *Nano Res.*, 2016, **9**, 435-441.
23 Z. J. Gu, T. Y. Zhai, B. F. Gao, X. H. Sheng, Y. B. Wang, H. B. Fu, Y. Ma and J. N. Yao, *J Phys. Chem. B*, 2006, **110**, 23829-23836.
24 A. Yella, U. K. Gautam, E. Mugnaioli, M. Panthoefer, Y. Bando, D. Golberg, U. Kolb and W. Tremel, *Crystengcomm*, 2011, **13**, 4074-4081.
25 M. Shibuya and M. Miyauchi, *Adv. Mater.*, 2009, **21**, 1373-1376.
26 W. B. Hu, Y. M. Zhao, Z. L. Liu, C. W. Dunnill, D. H. Gregory and Y. Q. Zhu, *Chem. Mater.*, 2008, **20**, 5657-5665.
27 D. D. Xu, T. F. Jiang, D. J. Wang, L. P. Chen, L. J. Zhang, Z. Fu, L. L. Wang and T. F. Xie, *ACS Appl. Mater. Inter.*, 2014, **6**, 9321-9327.
28 N. Li, Y. Zhao, Y. Wang, Y. Lu, Y. H. Song, Z. F. Huang, Y. W. Li and J. Z. Zhao, *Eur. J. Inorg. Chem.*, 2015, **17**, 2804-2812.
29 Y. S. Li, Z. L. Tang, J. Y. Zhang and Z. T. Zhang, *CrystEngComm*, 2015, **17**, 9102-9110.
30 S. Salmaoui and F. Sediri, *J. Alloy. Compd.*, 2016, **661**, 82-91.
31 S. Rajagopal, D. Nataraj, D. Mangalaraj, Y. Djaoued, J. Robichaud and O. Y. Khyzhun, *Nanoscale Res. Lett.*, 2009, **4**, 1335-1342.
32 K. Sato, M. Aoki and R. Noyori, *Science*, 1998, **281**, 1646-1647.
33 R. Noyori, M. Aoki and K. Sato, *Chem. Commun.*, 2003, **16**, 1977-1986.
34 M. Vafaeezadeh, M. M. Hashemi and M. Shakourian-Fard, *Catal. Commun.*, 2012, **26**, 54-57.
35 M. Vafaeezadeh and M. Mahmoodi Hashemi, *Chem. Eng. J.*, 2013, **221**, 254-257.
36 Y. Q. Wen, X. Y. Wang, H. J. Wei, B. J. Li, P. Jin and L. M. Li, *Green. Chem.*, 2012, **14**, 2868-2875.
37 M. Vafaeezadeh and M. Mahmoodi Hashemi, *Catal. Commun.*, 2014, **43**, 169-172.
38 M. Vafaeezadeh and M. Mahmoodi Hashemi, *RSC Adv.*, 2015, **5**, 31298-31302.
39 Z. H. Yang, Y. M. Yang, X. Zhu, G. D. Chen and W. X. Zhang, *Ind. Eng. Chem. Res.*, 2014, **53**, 9608-9615.
40 Z. H. Yang, Y. C. Zhang, W. X. Zhang, X. Wang, Y. T. Qian, X. G. Wen and S. H. Yang, *J. Solid. State. Chem.*, 2006, **179**, 679-684.
41 L. Seguin, M. Figlarz and J. Pannetier, *J. Alloy. Compd.*, 1993, **63**, 437-441.
42 K. De Buysser, I. Van Driessche, P. Vermeir, T. T. Thuy, J. Schaubroeck and S. Hoste, *Phys. Status. Solidi. B*, 2008, **245**, 2483-2489.
43 A. Chemseddine and U. Bloeck, *J. Solid. State. Chem.*, 2008, **181**, 2731-2736.
44 V. B. Patil, P. V. Adhyapak, S. S. Suryavanshi and I. S. Mulla, *J. Alloy. Compd.*, 2014, **590**, 283-288.
45 J. F. Yin, H. Q. Cao, J. X. Zhang, M. Z. Qu and Z. F. Zhou, *Cryst. Growth Des.*, 2013, **13**, 759-769.
46 C. Wang, R. Z. Sun, X. Li, Y. F. Sun, P. Sun, F. M. Liu and G. Y. Lu, *Sensors and Actuators B: Chemical*, 2014, **204**, 224-230.
47 Z. H. Yang, Y. M. Pan, Z. S. Mei and W. X. Zhang, *Appl. Surf. Sci.*, 2012, **258**, 4756-4763.
48 W. X. Zhang, H. Wang, Z. H. Yang and F. Wang, *Colloid Surface A*, 2007, **304**, 60-66.
49 Q. G. Ma, W. Z. Xing, J. H. Xu and X. H. Peng, *Catal. Commun.*, 2014, **53**, 5-8.
50 Y. Usui and K. Sato, *Green. Chem.*, 2003, **5**, 373-375

View Article Online
DOI: 10.1039/C6CE01940D

Table of Contents Graphic

Tungsten oxide three-dimensional hierarchical nanostructures with well-defined morphologies were synthesized via a facile pH-controlled approach, which show promising catalytic properties.